

AI Styling Studio - Feature Documentation

Overview

The **AI Styling Studio** is an MVP feature built into the Admin Consultation Dashboard of The Hideaway. It empowers stylists and administrators to generate AI-driven visual analyses of a client's portrait. Instead of manual mockups, the system leverages locally-run AI image generation models to produce high-quality, side-by-side comparisons of hairstyles and makeup variations, which are then sent to the client via the consultation chat.

UX Flow

- Trigger:** The Admin navigates to a specific pending or approved consultation in the Dashboard (`/admin/consultations/{id}`).
- Action:** In the left-hand panel under "AI Studio", the Admin selects:
 - Generate Hairstyle Options
 - Generate Makeup Options
 - (Optional Future Expansion: Personal Color Analysis)
- Processing:** The system takes the client's `CURRENT_HAIR` image (base64) and sends it to a local AI Image Generation model alongside a strictly defined system prompt.
- Delivery:** The generated infographic/variation image is uploaded to the Supabase `consultation_images` storage bucket.
- Presentation:** A new message is automatically inserted into the chat as the "Stylist", embedding the generated image for the client to review.

Core Prompts (Optimized for Local/Image Generation)

To ensure the local AI model generates clean, "visual-first" consultancy reports without unnecessary paragraphs of text, we utilize strict prompts. After testing with image generation, we have optimized the prompt to ensure better grid layouts and side-by-side comparisons.

1. Hairstyle Analysis

- Purpose:** Shows side-by-side hairstyle variations on the client's face shape.
- Mockup:**

Hairstyle Analysis

2. Visual Makeup Analysis

- Purpose:** Suggests makeup looks that complement the new hair color.
- Mockup:**

Makeup Analysis

3. Personal Color Analysis

- Purpose:** Identifies the user's seasonal color palette to guide overall styling.
- Mockup:**

Color Analysis

Implementation Status

- [x] Admin UI Integration (Sidebar Buttons)
- [x] Mockup Generation Workflow
- [x] Supabase Chat Integration
- [] Local AI Model Integration (SDAPI/Fooocus)

Technical Architecture (MVP)

1. Frontend Integration

- Component:** `app/admin/consultations/{id}/page.tsx`
- UI Elements:** Action buttons located in the AI Studio section.
- State Management:** Buttons will feature a loading spinner and disable state while awaiting the local API response to prevent duplicate generations.

2. Backend Pipeline

- **Endpoint:** POST /api/ai/generate-visuals
- **Payload:** { consultationId: string, type: 'hairstyle' | 'makeup', imageUrl: string }
- **Logic:**
 1. Fetch the image from the provided URL and convert it to the format required by the local model (e.g., base64).
 2. Send the image and the corresponding prompt to the LOCAL_IMAGE_MODEL_URL (e.g., Stable Diffusion API endpoint).
 3. Receive the generated image blob/base64.
 4. Upload the generated image to Supabase Storage (consultation_images).
 5. Insert a new record into the consultation_messages table referencing the new image URL, attributed to the stylist.

3. Environment Variables

To ensure the feature can run flexibly across different local setups without code changes, the following environment variables will be required:

```
# URL for the locally running image model (e.g., Automatic1111, ComfyUI)
LOCAL_IMAGE_MODEL_URL=http://localhost:7860/sdapi/v1/img2img

# Authentication token for the local API (if required)
LOCAL_IMAGE_MODEL_TOKEN=your_token_here
```

Future Considerations

- **Model Tuning:** Local models may require specific ControlNet configurations (e.g., IP-Adapter or FaceID) to ensure the client's facial identity is perfectly preserved across generated variations.
- **Webhooks:** If generation times exceed standard Vercel/Next.js API timeout limits (usually 10s-60s), the architecture may need to be upgraded to an asynchronous polling or webhook model.